GUJARAT STATE LOAD DESPATCH CENTRE

Security Constrained Unit Commitment & Economic Dispatch

SCUC / SCED Optimization Model

Technical Documentation & Model Report

Description of Model
Developed a Security Constrained Unit Commitment & Economic Dispatch (SCUC/SCED) optimization model for the Gujarat State Grid using GAMS, GAMSPy, and CPLEX solver. Modelled 47 generators across 5 transmission zones over 96 × 15-minute scheduling intervals (full 24-hour horizon). Implemented 10 constraint families including MUT/MDT, ramp rates, spinning reserve, inter-zonal power flows, and RE integration. Solved a 32,406-row, 13,489-binary MIP to determine optimal daily dispatch cost of Rs 122.4 Crore with <0.1% optimality gap. Produced 6 publication-quality visualizations including dispatch stack, merit order curve, commitment heatmap, and zonal analysis charts. Tools: GAMS 53.5.1, GAMSPy 1.23.1, CPLEX, Python 3.13, Pandas, Matplotlib.

1. Executive Summary

This document presents a Security Constrained Unit Commitment and Economic Dispatch (SCUC/SCED) optimization model developed for the Gujarat State Load Despatch Centre (SLDC) grid. The model determines the least-cost 24-hour generation schedule for 47 power generating units across 5 transmission zones, subject to technical, operational, and network constraints.

Key Result
Optimal daily operating cost: Rs 1,22,37,33,408 (Rs 122.4 Crore) for a peak demand of 24,400 MW. Average variable cost Rs 3.82/kWh. Model solved to 0.1% optimality gap using CPLEX solver in under 3 minutes.

1.1 What This Model Does

The model answers the core question every grid operator faces every day:

- Which power plants should be started up and shut down over the next 24 hours?
- How much power should each running plant generate in each 15-minute block?
- How much power should flow between zones to balance supply and demand?
- What is the minimum cost to reliably serve the entire state demand?

1.2 Model Scale

Generators	Time Blocks	Zones	MIP Variables
47 units	**96 × 15-min**	**5 zones**	**13,489 binary**
Coal, Gas, Lignite Hydro, Nuclear, Biomass	Full 24 hours (00:00 – 23:45 IST)	Kutch, Saurashtra N.Guj, Central, S.Guj	32,406 rows 19,442 columns

2. Data Description

2.1 What Day Is This Data From?

Important clarification: This model uses a **proxy dataset representing a typical Gujarat summer day**. It is NOT actual SCADA/metered data from a specific calendar date. The data has been constructed from the following authoritative sources:

Data Category	Source / Basis	Authenticity
Generator Capacities (Pmax/Pmin)	CEA Installed Capacity Reports 2024–25; CERC Tariff Orders	**Real generator data**
Variable Costs (VC)	CERC Annual Fixed/Variable Charge Orders; Merit Order Stack POSOCO	**Based on real tariffs**
Ramp Rates	IEGC (Indian Electricity Grid Code) 2010 — Clause 5.2 (MW/min norms)	**IEGC standard norms**
MUT / MDT	Standard thermal plant operation norms; CERC RE Integration Guidelines	**Industry standard**
24-hour Demand Profile	Gujarat SLDC published load data (typical summer 2024 curve, ~24,400 MW peak)	**Based on published data**
Solar Profiles (5 zones)	NRSC/IMD solar irradiance maps for Gujarat; GERMI solar resource atlas	**Zone-wise realistic**
Wind Profiles	Gujarat Energy Research and Management Institute (GERMI); Kutch & Saurashtra wind corridors	**Corridor-specific**
Transfer Limits	Gujarat Transmission Corporation (GETCO) 400kV/220kV corridor capacities; PGCIL planning data	**Approximate (proxy)**
Zonal Demand Split	Estimated from Gujarat SLDC zone-wise load reports (12/18/14/35/21% shares)	**Estimated (not metered)**

3. Mathematical Model Formulation

3.1 Problem Type

This is a Mixed Integer Linear Program (MILP) — also called Mixed Integer Programming (MIP). It combines:

- Continuous variables: generation dispatch (MW), power flows (MW)
- Binary (0/1) variables: unit commitment (on/off), startup events, shutdown events

MILP problems are NP-hard — solving them exactly requires sophisticated branch-and-bound solvers like CPLEX. The computational complexity is why grid operators need specialized software.

3.2 Objective Function

Minimize Total Operating Cost
TotalCost = Sum[VC(i) × 250 × P(i,t)] + Sum[NoLoad(i) × 0.25 × u(i,t)] + Sum[SU_Cost(i) × su(i,t)] + Sum[VOLL × 0.25 × USE(z,t)] Where: VC = variable cost (Rs/kWh), 250 = unit conversion (0.25hr × 1000kW/MW), NoLoad = no-load cost (Rs/hr), SU_Cost = startup cost (Rs/event), VOLL = Rs 50,000/MWh (Value of Lost Load — CERC standard)

3.3 Decision Variables

Variable	Type	Domain	Description
P(i,t)	Continuous	[0, Pmax(i)]	Dispatch MW of generator i in block t
u(i,t)	Binary {0,1}	{0, 1}	Commitment status: 1=online, 0=offline
su(i,t)	Binary {0,1}	{0, 1}	Startup event: 1=starts at block t
sd(i,t)	Binary {0,1}	{0, 1}	Shutdown event: 1=shuts down at block t
Flow(z,zz,t)	Continuous	[-TL, +TL]	Inter-zonal power flow MW (signed)
Curtail(z,t)	Continuous	[0, RE_Avail]	RE curtailment MW in zone z, block t
USE(z,t)	Continuous	[0, Demand]	Unserved energy slack variable (Rs 50,000/MWh penalty)

3.4 Constraints

#	Constraint	Equation Name	Mathematical Form	Count
1	**Power Balance**	*eq_balance(z,t)*	Generation + RE - Curtailment + Net Import = Demand	480
2	**Maximum Generation**	*eq_pmax(i,t)*	P(i,t) ≤ Pmax(i) × u(i,t)	4,512

#	Constraint	Equation Name	Mathematical Form	Count
3	**Minimum Generation**	*eq_pmin(i,t)*	P(i,t) ≥ Pmin(i) × u(i,t)	4,512
4	**Ramp Up Limit**	*eq_ramp_up(i,t)*	P(i,t) - P(i,t-1) ≤ RampUp(i)	4,465
5	**Ramp Down Limit**	*eq_ramp_dn(i,t)*	P(i,t-1) - P(i,t) ≤ RampDn(i)	4,465
6	**Startup/Shutdown Logic**	*eq_su_sd(i,t)*	su(i,t) - sd(i,t) = u(i,t) - u(i,t-1)	4,465
7	**Minimum Up Time**	*eq_mut(i,t)*	Sum of startups in MUT window ≤ u(i,t)	4,465
8	**Minimum Down Time**	*eq_mdt(i,t)*	Sum of shutdowns in MDT window ≤ 1 - u(i,t)	4,465
9	**Spinning Reserve**	*eq_spin(t)*	Sum[Pmax×u] - Sum[P] ≥ 5% × Demand(t)	96
10	**Flow Anti-symmetry**	*eq_antisym(z,zz,t)*	Flow(z,zz,t) + Flow(zz,z,t) = 0	480

3.5 Network Structure

Zone	Demand Share	Avg Demand	Key Generators	RE Sources
KUTCH	12%	~2,413 MW avg	PANANDHRO, BECL, EPGL, APL MUNDRA, CGPL, ACB, TP AECO	Solar + Wind (highest RE)
SAURASHTRA	18%	~3,620 MW avg	SIKKA TPS, GSPC PIPAVAV	Wind (SAUR corridor)
N_GUJ	14%	~2,816 MW avg	None (import-dependent zone)	Solar (N Gujarat)
CENTRAL	35%	~7,040 MW avg	GANDHINAGAR, WANAKBORI, KADANA, VSTPS, KORBA, SIPAT, NTPC, SSP, BIOMASS	Wind + Solar
S_GUJ	21%	~4,224 MW avg	UKAI, DHUVARAN, SUGEN, KAKRAPAR, TARAPUR, GIPCL, UKAI HYDRO	Solar (S Gujarat)

4. Generator Portfolio

Fuel Type	Units	Generators	Total Capacity (MW)
Coal	14	UKAI, WANAKBORI, GANDHINAGAR, SIPAT, VSTPS (I-V), KORBA, APL MUNDRA, CGPL, MAUDA, LARA, KHARGONE, DADRI, DB POWER	~18,200
Gas/CCPP	12	SUGEN, DHUVARAN, UTRAN II, KAWAS, JHANOR, EPGL, GSEG, GSPC, SLPP, GIPCL, UNOSUGEN, PIPAVAV	~6,700
Lignite	5	PANANDHRO, BECL, AKRIMOTA, TP AECO, ACB INDIA	~1,460
Hydro	4	UKAI HYDRO, KADANA, SSP CHPH, SSP RBPH	~779
Nuclear	4	KAKRAPAR KAPS 1&2, KAKRAPAR EXT 3&4, TARAPUR 1&2, TARAPUR 3&4	~1,035
Biomass	1	BIOMASS (CENTRAL zone)	~82

4.1 Must-Run Units

The following 6 units are constrained to remain online throughout all 96 blocks:

Unit	Pmax (MW)	Type	Reason
KAKRAPAR KAPS 1&2	125	Nuclear	*Nuclear cannot cycle — safety & economics*
KAKRAPAR EXT 3&4	475.88	Nuclear	*Nuclear cannot cycle — safety & economics*
TARAPUR 1&2	160	Nuclear	*Nuclear cannot cycle — safety & economics*
TARAPUR 3&4	274	Nuclear	*Nuclear cannot cycle — safety & economics*
SSP CHPH	40	Run-of-river Hydro	*Sardar Sarovar — must-run for irrigation*
SSP RBPH	192	Pumped Hydro	*Sardar Sarovar — grid stability obligation*

5. Optimization Results

Metric	Value	Notes
Optimal Total Cost	**Rs 1,22,37,33,408**	*~Rs 122.4 Crore per day*
Average Variable Cost	**Rs 3.82 / kWh**	*CERC benchmark: Rs 2–7/kWh ✓*
Peak System Demand	**24,400 MW**	*Block T81 (20:15 IST)*
Minimum System Demand	**19,304 MW**	*Block T52 (13:00 IST)*
Total Energy Dispatched	**1,85,286 MWh**	*Thermal + Hydro + Nuclear*
RE Generation (Solar+Wind)	**~80,000 MWh**	*Fully absorbed, zero curtailment*
Units Committed (min)	**32 units**	*Block T46 (11:30 IST)*
Units Committed (max)	**43 units**	*Block T79 (19:45 IST)*
Units Committed (avg)	**37.1 units**	*Out of 47 total*
Spinning Reserve	**Satisfied all 96 blocks**	*Min 5% of demand maintained*
Optimality Gap	**≤ 0.1%**	*CPLEX solver, 7200s time limit*
Power Balance Accuracy	**99.97%**	*51.7 MWh residual (physical limit)*
Highest Cost Generator	**APL ADANI MUNDRA**	*Rs 17.3 Crore (Coal, KUTCH zone)*
Lowest Cost Generator	**KORBA NTPC**	*Rs 1.37/kWh variable cost*

5.1 Merit Order Insights

The model dispatches generators strictly in merit order (cheapest first), subject to constraints:

- Cheapest: KORBA NTPC (Rs 1.27–1.39/kWh) — runs at maximum capacity throughout
- Mid-range: Coal plants (Rs 2.1–5.5/kWh) — committed and dispatched during peak hours
- Most expensive: Gas peakers DHUVARAN CCPP (Rs 13.08/kWh), UTRAN II (Rs 12.73/kWh) — only committed during peak demand T70–T85
- Nuclear & Hydro: Always online (must-run) at lowest variable cost (Rs 2.05–4.40/kWh)

5.2 RE Integration

Solar and wind generation is modeled as must-take (zero variable cost, priority dispatch):

- Peak solar: ~9,800 MW at T49 (12:15 IST) — primarily from KUTCH (1,796 MW) and N_GUJ (3,493 MW) zones
- Wind generation: 4,248 MW at midnight, dropping to 2,720 MW at noon, recovering to 5,206 MW in evening
- Zero RE curtailment: all solar and wind absorbed in every block — validates adequate thermal flexibility

6. Model Validation

Validation Test	Result	Details	Status	Value
Power Balance Accuracy	99.97%	*Unserved energy: 51.7 MWh / 185,286 MWh total*	**PASS**	
Pmin / Pmax Limits	0 violations	*All 47 units × 96 blocks satisfied*	**PASS**	
Ramp Rate Constraints	0 violations	*All start-up/ramp transitions valid*	**PASS**	
Spinning Reserve	0 shortfalls	*≥5% of demand in all 96 blocks*	**PASS**	
Must-Run Units	100% online	*Nuclear + SSP hydro committed throughout*	**PASS**	
RE Curtailment	0 MWh	*All solar & wind generation absorbed*	**INFO**	
Cost Sanity	Rs 3.82/kWh	*CERC benchmark range: Rs 2–7/kWh*	**PASS**	
Units Online	32–43 / block	*Average 37.1 units per 15-min block*	**PASS**	
Residual Imbalance	51.7 MWh	*S_GUJ T80–T81, KUTCH T88 (peak hours)*	**NOTE**	

6.1 Residual Imbalance — Explanation

Why 51.7 MWh residual gap is acceptable — and honest

Blocks S_GUJ T80 (84 MW), S_GUJ T81 (57 MW), KUTCH T88 (65 MW) show a small supply deficit during peak demand hours (20:00–22:00 IST). This is NOT a model error — it is the correct physical outcome: at peak, S_GUJ generators are at Pmax, CENTRAL-to-S_GUJ transfer is at its limit (2,800 MW), and no additional supply is physically available without violating other constraints. The 51.7 MWh gap represents 0.03% of total daily energy — well within the 0.1% threshold used in academic literature. Artificially inflating transfer limits to eliminate this gap would make the model less realistic, not more. This is the grid telling us: South Gujarat needs more local generation or transmission augmentation during evening peak.

7. Tools & Technology Stack

Component	Tool / Version	Purpose
Optimization Language	GAMS 53.5.1 + GAMSPy 1.23.1	Industry standard for power systems MIP
MIP Solver	CPLEX (IBM)	World's best commercial MIP solver
Programming Language	Python 3.13	Model building, data processing, visualization
Data Processing	Pandas + NumPy	Parameter extraction, results analysis
Visualization	Matplotlib	6 publication-quality figures
Data Source	Excel (SLDC format)	47 generators, 5 zones, 96 time blocks
Output Format	Excel (.xlsx)	9 sheets: dispatch, commitment, costs, flows
Licence	GAMSPy Academic Network	GPA112384

7.1 Software Architecture

The model follows a 3-phase pipeline:

1. Phase 1 — Data Layer: Excel input → GAMS parameter blocks (47 generators, 96 blocks, 5 zones, RE profiles, demand curves)
2. Phase 2 — Optimization Layer: GAMSPy Python API builds the MIP model → CPLEX solver finds optimal solution → Results exported to Excel (9 sheets)
3. Phase 3 — Visualization Layer: Python/Matplotlib generates 6 publication-quality figures (300 DPI, white background, journal standard)

7.2 Output Files

- Gujarat_SCUC_Results.xlsx — 9 sheets: Dispatch_MW, Commitment, Startups, System_Summary, Unit_Cost_Summary, Zonal_Flows, RE_Curtailment, Full_Results, Unserved_Energy
- Fig1_Dispatch_Stack.png — 24-hour generation stack chart
- Fig2_Commitment_Heatmap.png — 47×96 commitment schedule heatmap
- Fig3_Merit_Order.png — merit order curve with demand intersections
- Fig4_Cost_Breakdown.png — top 15 generators + fuel-wise cost pie
- Fig5_System_Metrics.png — load-generation balance + spinning reserve
- Fig6_Zonal_Analysis.png — 5-zone supply/demand with model summary

8. Limitations & Future Work

Being honest about model limitations is as important as reporting results. The following gaps exist between this model and a full production-grade SCUC/SCED system:

Gap	Priority	Description	How to Fix
N-1 Security Constraints	High	Real SCUC checks if grid is stable after any single line trip. Our model uses fixed transfer limits.	*Add PTDF matrix, contingency constraints*
Real Meter Data	High	Zonal demand uses fixed ratio (12/18/14/35/21%). Real SLDC uses SCADA meter data.	*Integrate SLDC portal historical data*
Hydro Reservoir Model	Medium	Hydro treated like thermal. Real model needs water availability, irrigation constraints.	*Add energy budget & reservoir level constraints*
Central Sector (ISGS)	Medium	NTPC/nuclear allocations are free variables. Real model has pre-fixed state shares.	*Fix ISGS allocations from NLDC daily schedule*
Startup Time (Hot/Warm/Cold)	Medium	Assumes instant startup. Real thermal units need 2–8 hours.	*Add startup trajectory constraints*
AC Power Flow	Low	Uses DC approximation (transfer limits). Real SCUC uses full AC load flow.	*Integrate Newton-Raphson AC power flow*
Ancillary Services	Low	No frequency regulation or reactive power. Simplified spinning reserve only.	*Add AGC, frequency response constraints*
Bilateral/PPA Contracts	Low	Long-term PPAs not pre-fixed. Minor impact on dispatch order.	*Fix PPA commitments as must-run with price caps*
Emission Constraints	Low	No CO_2/SO_2 limits. Growing importance for compliance.	*Add emission caps per generator/fuel type*
Transmission Losses	Low	Lossless network assumed. Real losses ~3–4% zonal.	*Add loss factor multipliers per zone pair*

8.1 Priority Improvement Roadmap

4. N-1 Security Constraints — Add DC Power Flow (PTDF matrix). Estimated effort: 2–3 weeks
5. Real SCADA Data — Integrate Gujarat SLDC portal API for actual 15-min metered demand. Estimated effort: 1 week
6. Hydro Reservoir Model — Add water availability and energy budget constraints. Estimated effort: 1 week
7. ISGS Central Sector Fix — Pre-fix NTPC/nuclear state allocations from NLDC daily schedules. Estimated effort: 3 days
8. Startup Trajectories — Model hot/warm/cold start with different cost and time parameters. Estimated effort: 1 week

End of Document

Gujarat SLDC SCUC/SCED Model | Surjeet Chauhan | June 2026